

GEOSTEER: Geodesic Optimization for Activation Steering in Large Language Models

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Abstract

Activation steering provides a lightweight way to control large language models (LLMs) by modifying their hidden activations at inference time. Among these approaches, norm-preserving steering aims to change model behavior without altering the activation norm, reducing the risk of representation collapse and degradation. However, existing norm-preserving methods are limited by predefined steering trajectories and by their reliance on one-step updates, which may fail to capture the complex structure of activation distributions. We propose **GEOSTEER**, an optimization-based method for *norm-preserving activation steering*. GEOSTEER formulates steering as a Riemannian optimization problem and updates activations through a sequence of small geodesic steps on the representation manifold. To avoid fixed steering directions, GEOSTEER learns a nonlinear activation-space objective that distinguishes desired from undesired activations, and uses this function to adaptively guide each steering step. This multistep formulation yields smoother, more stable, and more consistent steering behavior while preserving the activation norm. Across TruthfulQA, RealToxicityPrompts, and UltraFeedback benchmarks, GEOSTEER consistently improves over state-of-the-art activation steering baselines. These results suggest that norm-preserving steering can be made more effective by replacing predefined one-step edits with adaptive, geometry-aware optimization.

societal impacts, including hallucination, bias, toxicity, and other harmful behaviors. Since modern LLMs often contain hundreds of billions of parameters (Chowdhery et al., 2023; Brown et al., 2020), controlling and aligning them through fine-tuning is becoming increasingly challenging and resource-intensive. *This motivates the need for lightweight alternatives that can adapt model behavior without updating model parameters.*

Activation steering, or representation engineering, offers a lightweight way to influence LLM behavior by modifying hidden states at inference time (Rimsky et al., 2024; Wehner et al., 2025; Bartoszcze et al., 2025). A common strategy is to construct a steering vector from contrastive examples and add it to model activations during generation (Li et al., 2023; Rimsky et al., 2024; Liu et al., 2023; Lee et al., 2024). While effective in many settings, such additive edits are highly sensitive to the intervention scale. If the intervention is too weak, the model behavior changes little; if it is too strong, the edited activation may move away from the model’s natural representation space and harm generation quality (You et al., 2026). This limitation has motivated recent work on more structured intervention mechanisms, including learnable and geometry-aware approaches (Wu et al., 2024; Deng et al., 2026). At the same time, the architecture of modern LLMs suggests a useful geometric perspective: normalization layers such as RMSNorm (Zhang and Sennrich, 2019) regulate activation scale, making the direction of the hidden representation an important target for controlled steering.

Norm-preserving steering methods have therefore been proposed as a way to control model behavior while maintaining the activation norm (Pham and Nguyen, 2024; You et al., 2026; Vu and Nguyen, 2026). However, existing norm-preserving methods still have important limitations, as conceptually illustrated in Figure 1. While they

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities and have become an essential component of modern AI, supporting a wide range of applications (Achiam et al., 2023; Dubey et al., 2024; Liu et al., 2024; Yang et al., 2025). However, as LLMs continue to develop rapidly, it is increasingly important to study and control their

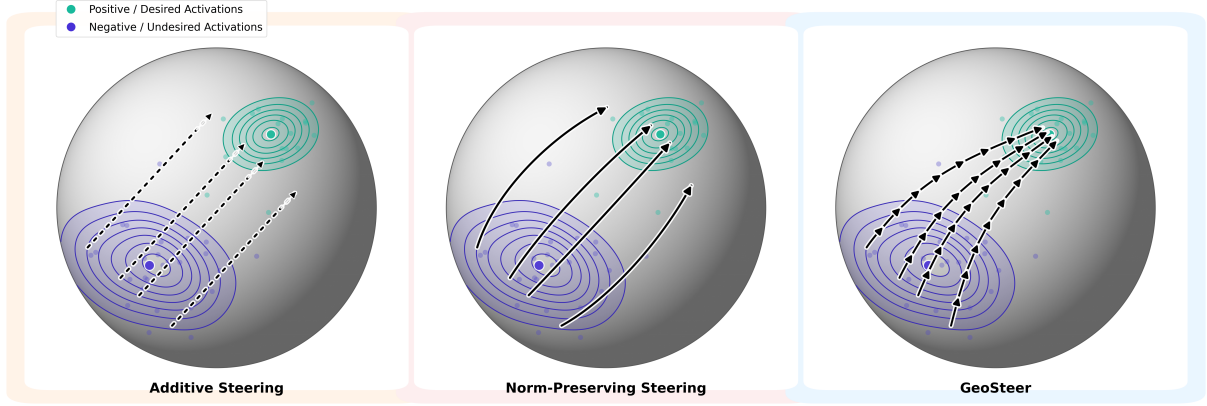


Figure 1: Conceptual illustration of GEOSTEER. Additive steering can move activations off the sphere, while existing norm-preserving steering keeps activations on the sphere but typically follows predefined or heuristic trajectories. In contrast, GEOSTEER adaptively optimizes the steering path by decomposing the intervention into multiple geodesic updates. This produces smoother trajectories toward the desired activation region while preserving the activation norm.

avoid the unconstrained Euclidean shifts used by additive steering, they often follow predefined or heuristic trajectories rather than adaptively optimizing the steering path. For example, one-step approaches such as Angular Steering (Vu and Nguyen, 2026) and Spherical Steering (You et al., 2026) typically rely on simple statistical features, such as mean activation differences, while ignoring richer distributional structure and interactions among activation dimensions. Their reliance on one-step updates may also fail to capture the complex patterns of activation distributions, limiting their expressive power when steering nuanced model behaviors.

Other methods, such as HPR (Pham and Nguyen, 2024), introduce more complex training procedures involving neural networks, but still assume that the correct steering direction lies within a predefined two-dimensional plane. This geometric restriction can limit flexibility, especially when the desired behavioral change requires movement along more complex directions in representation space. Moreover, existing methods are often sensitive to hyperparameters and may not generalize consistently across different models or tasks.

To address these limitations, we propose GEOSTEER, an optimization-based framework for norm-preserving activation steering. We observe that norm-preserving steering can be naturally viewed as moving an activation point on a Riemannian manifold from an undesired region toward a desired region, while keeping its norm fixed. Rather than applying a single large update or following a predefined rotation, GEOSTEER decom-

poses steering into a sequence of small geodesic updates (Absil et al., 2008). Each update is guided by the current activation state, and together these steps form a smooth, adaptive trajectory on the manifold. We formulate this process as a Riemannian optimization problem (Absil et al., 2008). Specifically, we define a nonlinear activation-space objective that distinguishes desired from undesired activations. This objective provides an adaptive signal for selecting steering directions that increase alignment with desired activations while moving away from undesired ones. At inference time, GEOSTEER performs iterative geodesic updates guided by this objective, ensuring that activations remain on the norm-preserving manifold throughout the steering process. By replacing predefined one-step rotations with adaptive multistep optimization, GEOSTEER captures richer and more fine-grained structure in the activation space. This leads to more stable and expressive steering while preserving activation norms. Experiments on TruthfulQA (Lin et al., 2021), UltraFeedback (Cui et al., 2023), and Real-ToxicityPrompts (Gehman et al., 2020) show that GEOSTEER consistently outperforms state-of-the-art activation steering baselines.

Our contributions are summarized as follows:

- We formulate norm-preserving activation steering as Riemannian optimization on the activation sphere, providing a principled geometric view of activation intervention.
- We propose GEOSTEER, an adaptive multistep steering method that replaces fixed one-step transformations with a sequence of small

geodesic updates guided by the current activation state.

- We conduct experiments on TruthfulQA, UltraFeedback, and RealToxicityPrompts, showing that GEOSTEER consistently improves over strong activation steering baselines while maintaining efficient inference.

2 Related Work

Activation Steering. Activation steering controls LLM behavior by modifying hidden activations at inference time. Most existing methods follow an additive steering paradigm, where a direction vector is estimated from contrastive activation patterns and added to the model’s hidden states during decoding. The intervention typically takes the form of shifting an activation by a scaled steering vector, so that the edited representation is moved toward a desired behavioral direction in activation space. Fixed-vector methods such as ITI (Li et al., 2023), RepE (Zou et al., 2023), and CAA (Rimsky et al., 2024) apply the same update across inputs, which limits their adaptability. MiMiC (Singh et al., 2024) and Linear-AcT (Rodriguez et al., 2025) introduce more structured linear formulations, while ODESteer (Zhao et al., 2026) further improves adaptivity by treating steering as an ODE-based trajectory. *Nevertheless, these approaches do not explicitly constrain the activation norm, so their updates may still push representations away from the model’s natural activation geometry and degrade utility.*

Norm-Preserving Activation Steering. Norm-preserving activation steering addresses this issue by modifying activation directions while keeping activation norms unchanged. These methods are motivated by the observation that modern LLM activations often lie on structured manifolds, where changing the direction can influence model behavior without disrupting activation magnitude. Angular Steering (Vu and Nguyen, 2026) and HPR (Pham and Nguyen, 2024) perform geometric updates using fixed low-dimensional planes or Householder reflection, but they still impose strong assumptions on the steering direction. Spherical Steering (You et al., 2026) rotates activations on the sphere, but often relies on simple mean-based statistics and ignores richer distributional structure. Moreover, most norm-preserving methods perform only a one-step update, limiting their ability to capture complex activation patterns. *In contrast, our*

method performs adaptive multistep updates on the manifold, recomputing the steering direction at each step based on the current activation.

3 Methodology

3.1 Preliminaries

Activation Steering. Activation steering aims to modify an internal hidden representation of a pre-trained language model so that the model behavior shifts toward a desired attribute while preserving the original model parameters. Let $h \in \mathbb{R}^d$ denote the activation at a selected layer and token position. Conventional steering methods often edit the activation by adding a direction vector,

$$\hat{h} = h + \alpha v, \quad (1)$$

where v is a steering direction and α controls the steering strength. While simple, this additive update may change both the direction and norm of the activation, which can move the representation away from the geometry learned by the model.

Norm-Preserving Steering on the Hypersphere. Norm-preserving steering constrains the edited activation to remain on a hypersphere with the same norm as the original activation. Given an activation h , its normalized representation can be written as

$$z = \frac{h}{\|h\|_2}, \quad z \in \mathbb{S}^{d-1}. \quad (2)$$

Steering can then be viewed as modifying z on the unit sphere to obtain an edited normalized activation $\hat{z} \in \mathbb{S}^{d-1}$. The final edited activation in the original activation space is recovered by restoring the original norm:

$$\hat{h} = \|h\|_2 \hat{z}. \quad (3)$$

Here, $\hat{z}$ denotes the steered direction on the unit sphere, while $\hat{h}$ denotes the final edited activation used by the model. This formulation preserves the activation norm and treats steering as a directional update over the sphere rather than as an unconstrained Euclidean shift.

Riemannian Optimization on the Sphere. Steering on the hypersphere can be viewed as a constrained optimization problem. Given a normalized activation $z \in \mathbb{S}^{d-1}$ and an objective function $\mathcal{L}(z)$, the goal is to update z so as to improve the objective while maintaining the constraint

$$\|z\|_2 = 1. \quad (4)$$

A standard Euclidean gradient step does not generally satisfy this constraint, since it may move the activation away from the sphere. Riemannian optimization addresses this by computing the update direction in the tangent space of the manifold.

For the unit sphere, the tangent space at z is

$$T_z \mathbb{S}^{d-1} = \{u \in \mathbb{R}^d : u^\top z = 0\}. \quad (5)$$

This space contains all directions that are locally valid on the sphere. Given the Euclidean gradient

$$g = \nabla_z \mathcal{L}(z), \quad (6)$$

we obtain the Riemannian gradient by removing the component of g that points in the radial direction:

$$g_{\mathcal{R}} = g - (g^\top z)z. \quad (7)$$

The resulting vector $g_{\mathcal{R}} \in T_z \mathbb{S}^{d-1}$ gives the steepest local change of the objective under the spherical constraint. After choosing a tangent update direction, the point can be moved along the sphere using a geodesic update. This provides the geometric basis for GEOSTEER, where $\mathcal{L}(z)$ is instantiated as a learned steering objective.

3.2 GEOSTEER

We consider inference-time activation steering for a pretrained language model. Given an input prompt x , let $h \in \mathbb{R}^d$ denote the activation at a selected layer and token position. GEOSTEER edits this activation by optimizing its normalized direction on the unit sphere. We denote the inference-time trajectory by $\{z^{(t)}\}_{t=0}^K$, initialized as

$$z^{(0)} = \frac{h}{\|h\|_2}. \quad (8)$$

After K geodesic steps, the edited activation is reconstructed as

$$\hat{h} = \|h\|_2 z^{(K)}, \quad (9)$$

so the original activation norm is preserved.

Learning the Steering Objective. GEOSTEER first learns a probe that distinguishes desired from undesired activations. Given labeled activations

$$\mathcal{D} = \{(h_i, y_i)\}_{i=1}^N, \quad (10)$$

where $y_i \in \{0, 1\}$ denotes the undesired or desired class, we normalize each training activation as

$$z_i = \frac{h_i}{\|h_i\|_2}. \quad (11)$$

The probe score is defined as:

$$s_\phi(z) = w^\top \phi(z) + b, \quad (12)$$

where $\phi(\cdot)$ is a nonlinear feature map, and w and b are learnable parameters. The desired-class probability is

$$p_\phi(z) = \sigma(s_\phi(z)). \quad (13)$$

The probe is trained with the binary classification loss

$$\mathcal{L}_{\text{probe}} = -\sum_i [y_i \log p_\phi(z_i) + (1 - y_i) \log(1 - p_\phi(z_i))]. \quad (14)$$

In our implementation, $\phi(\cdot)$ is instantiated with Polynomial Count Sketch (PCS) (Pham and Pagh, 2013):

$$\phi(z) = \text{PCS}(z). \quad (15)$$

This provides an efficient nonlinear representation without explicitly expanding high-dimensional polynomial features. After training, the probe is frozen and used to define the inference-time steering objective.

Geodesic Steering. At inference time, GEOSTEER steers the activation by maximizing the desired-class probability under the learned probe. Equivalently, we minimize

$$\mathcal{L}_{\text{steer}}(z) = -\log p_\phi(z). \quad (16)$$

Starting from $z^{(0)}$, GEOSTEER performs K geodesic steps with step size $\eta = T/K$, where T is the total steering strength. At each step t , we compute the Euclidean gradient

$$g^{(t)} = \nabla_{z^{(t)}} \mathcal{L}_{\text{steer}}(z^{(t)}), \quad (17)$$

and project it onto the tangent space of the sphere:

$$g_{\mathcal{R}}^{(t)} = g^{(t)} - \left((g^{(t)})^\top z^{(t)}\right) z^{(t)}. \quad (18)$$

The normalized descent direction is

$$u^{(t)} = -\frac{g_{\mathcal{R}}^{(t)}}{\|g_{\mathcal{R}}^{(t)}\|_2}. \quad (19)$$

GEOSTEER then updates the activation direction along the geodesic on the unit sphere (Absil et al., 2008):

$$z^{(t+1)} = \cos(\eta) z^{(t)} + \sin(\eta) u^{(t)}. \quad (20)$$

Because this update follows the sphere, each intermediate point remains on $\mathbb{S}^{d-1}$. We provide the proof in Appendix A.4. After K steps, the edited activation is recovered as

$$\hat{h} = \|h\|_2 z^{(K)}. \quad (21)$$

The full inference procedure is summarized in Algorithm 1.

Algorithm 1: GEOSTEER Inference

Require: Activation h , frozen probe p_ϕ , total steering strength T , number of steps K

Ensure: Steered activation $\hat{h}$

```
1:  $z^{(0)} \leftarrow h / \|h\|_2$ 
2:  $\eta \leftarrow T / K$ 
3: for  $t = 0$  to  $K - 1$  do
4:    $\mathcal{L}_{steer}(z^{(t)}) \leftarrow -\log p_\phi(z^{(t)})$ 
5:    $g^{(t)} \leftarrow \nabla_{z^{(t)}} \mathcal{L}_{steer}(z^{(t)})$ 
6:    $g_{\mathcal{R}}^{(t)} \leftarrow g^{(t)} - ((g^{(t)})^\top z^{(t)}) z^{(t)}$ 
7:    $u^{(t)} \leftarrow -g_{\mathcal{R}}^{(t)} / \|g_{\mathcal{R}}^{(t)}\|_2$ 
8:    $z^{(t+1)} \leftarrow \cos(\eta) z^{(t)} + \sin(\eta) u^{(t)}$ 
9: end for
10:  $\hat{h} \leftarrow \|h\|_2 z^{(K)}$ 
11: return  $\hat{h}$ 
```

3.3 Comparison with Existing Norm-Preserving Steering.

GEOSTEER differs from prior norm-preserving steering methods in two main aspects. First, rather than using a fixed geometric transformation, it treats activation steering as an optimization problem on the activation sphere. Second, the steering direction is determined by the local gradient of a nonlinear probe, allowing the direction to adapt at each step. This enables GEOSTEER to follow a curved trajectory toward the desired activation region, instead of relying on a single global direction or a fixed two-dimensional rotation plane for all activations.

4 Experiments

In this section, we evaluate GEOSTEER across three representative activation-steering tasks: helpfulness, truthfulness, and detoxification. These tasks test whether the proposed method can steer model behavior toward desirable attributes while preserving generation quality.

Base LLMs. We conduct experiments on four open-source language models with comparable parameter scales: Falcon-7B (Almazrouei et al., 2023), Mistral-7B-v0.3 (Jiang et al., 2023), LLaMA-3.1-8B (Meta AI, 2024), and Qwen2.5-7B (Team, 2024). Additional details on the model implementations are provided in Appendix A.5.

Baselines. We compare GEOSTEER against a diverse set of activation-steering baselines. These include Representation Engineering (RepE) (Zou et al., 2023), Inference-Time Intervention (ITI) (Li et al., 2023), Contrastive Activation Addition

(CAA) (Rimsky et al., 2024), Minimally Modified Counterfactuals (MiMiC) (Singh et al., 2024), Linear-AcT (Rodriguez et al., 2025), and ODESteer (Zhao et al., 2026). We also include norm-preserving methods that are most closely related to our setting, including HPR (Pham and Nguyen, 2024), Spherical Steering (You et al., 2026), and Angular Steering (Vu and Nguyen, 2026). To ensure a controlled comparison, all methods are evaluated under the same steering protocol: we use the same steering layer and apply the intervention to newly generated tokens, following common practice in prior activation-steering evaluations (Wehner et al., 2025; Bartoszcze et al., 2025). Additional details on the selected steering layers are provided in Appendix A.6.

Scope of Comparison. We focus on methods that are directly compatible with single-layer inference-time activation steering. Therefore, we do not include methods designed for substantially different goals, such as multi-attribute control (Nguyen et al., 2025), differential privacy (Goel et al., 2025), or instruction-following improvement (Stolfo et al., 2025). We also exclude SADI (Wang et al., 2025), as it requires interventions across all layers and therefore does not fit our single-layer evaluation setup.

Datasets and Metrics. We evaluate on benchmarks corresponding to the three steering objectives. For helpfulness, we use UltraFeedback (Cui et al., 2023) and report the win rate against responses from the original model as the main metric, along with mean reward and 90th-percentile reward. For truthfulness, we evaluate on TruthfulQA (Lin et al., 2021), using Truth $\times$ Info as the primary metric and reporting Truth and Info separately for completeness. For detoxification, we use RealToxicityPrompts (Gehman et al., 2020) and report toxicity as the main metric. We additionally report perplexity and Dist- n to measure generation quality and diversity. Further details on the datasets and evaluation metrics are provided in Appendix A.7.

4.1 Experimental Results

Table 1 reports the main results across helpfulness, truthfulness, and detoxification. Overall, GEOSTEER consistently achieves the best performance on the primary metrics across all evaluated models. Specifically, GEOSTEER obtains the highest win rate on UltraFeedback, the highest truthfulness $\times$ informativeness score on TruthfulQA, and the lowest toxicity score on RealToxi-

Table 1: Comparison across Falcon-7B, Mistral-7B, and LLaMA3.1-8B on helpfulness, truthfulness, and detoxification. For helpfulness, “Win” measures the win rate over the original model, and “RM_{mean}” / “RM_{P90}” report the mean and 90th-percentile reward. For truthfulness, “T×I” denotes Truthfulness × Informativeness, with “True” and “Info” shown separately. For detoxification, “PPL” denotes perplexity. Results are averaged over five runs. Primary metrics are highlighted in blue ; for main and selected auxiliary metrics, the best and second-best results are marked by **bold** and underline.

Method	Model	Helpfulness (Ultrafeedback)			Truthfulness (TruthfulQA)			Detoxification (Real Toxicity Prompts)		
		Win (%)↑	RM _{mean} ↑	RM _{P90} ↑	T×I (%)↑	True (%)↑	Info (%)↑	Toxic ↓	PPL↓	Dist-2↑
Original		50.0	-12.5	-2.9	28.4	31.3	96.3	0.181	15.8	94.4
RepE	Falcon-7B	49.7	-12.9	-2.8	28.7	31.6	96.2	0.187	16.3	94.7
ITI		48.5	-12.5	-2.9	29.4	32.5	96.1	0.187	15.6	94.5
CAA		<u>54.0</u>	-12.4	-1.2	34.1	37.6	96.0	0.093	17.1	94.7
MiMiC		45.5	-13.1	-3.4	35.4	43.9	90.1	0.167	16.5	94.1
Linear-AcT		49.8	-12.8	-2.5	34.1	37.7	96.1	0.168	15.9	94.6
ODESteer		49.0	-12.8	-3.4	33.3	36.2	96.6	<u>0.088</u>	37.5	93.4
HPR		48.9	-13.5	-3.5	34.9	39.2	95.7	0.099	43.1	92.9
Angular Steering		51.5	-12.5	-2.9	<u>39.4</u>	43.7	94.7	0.113	16.6	96.0
Spherical Steering		51.8	<u>-12.1</u>	-2.3	33.4	<u>45.2</u>	86.7	0.101	19.0	93.8
GEOSTEER (Ours)		57.3	-11.8	<u>-2.0</u>	41.1	46.5	93.6	0.071	17.9	93.5
Original		50.0	-11.3	-1.2	36.7	40.8	95.6	0.125	18.7	98.9
RepE	Mistral-7B	36.6	-14.5	-3.4	33.3	43.8	88.9	0.116	39.9	93.2
ITI		47.0	-11.2	-1.0	42.6	48.3	94.0	0.060	22.2	99.0
CAA		<u>53.4</u>	<u>-9.6</u>	0.6	53.2	58.6	94.5	0.041	20.8	99.0
MiMiC		49.0	-11.4	-0.8	40.0	47.1	92.7	0.103	18.9	98.9
Linear-AcT		51.1	-10.9	-0.1	42.6	47.6	95.0	0.102	19.3	98.9
ODESteer		53.3	-10.5	<u>0.7</u>	51.5	56.9	94.5	<u>0.034</u>	20.2	99.3
HPR		52.3	-10.1	0.7	50.4	56.4	89.4	0.058	43.1	92.9
Angular Steering		50.4	-11.1	-0.7	51.0	58.0	93.0	0.049	31.1	99.2
Spherical Steering		47.3	-11.9	-2.4	<u>58.1</u>	<u>69.9</u>	86.8	0.044	34.3	98.2
GEOSTEER (Ours)		54.7	-8.9	0.9	63.2	70.0	93.0	0.015	39.6	99.1
Original		50.0	-10.5	-0.3	41.5	44.1	96.7	0.124	18.7	99.0
RepE	LLaMA3.1-8B	49.4	-10.4	-1.2	40.6	43.3	96.7	0.143	38.5	96.4
ITI		48.8	-10.5	-0.5	43.7	46.1	96.9	0.087	19.7	98.6
CAA		57.6	-8.9	1.5	50.1	53.0	97.6	0.042	21.5	99.2
MiMiC		57.8	-9.2	1.1	54.1	60.8	92.5	0.098	19.2	99.3
Linear-AcT		52.8	-9.9	0.9	51.5	53.9	97.1	0.103	18.7	98.9
ODESteer		<u>59.0</u>	-8.5	1.6	<u>67.7</u>	72.6	94.7	0.032	24.6	98.7
HPR		52.2	-10.0	0.7	57.0	60.7	94.0	0.040	36.1	97.5
Angular Steering		55.7	<u>-8.2</u>	<u>1.7</u>	67.1	71.2	95.5	0.043	20.6	99.1
Spherical Steering		52.3	-9.5	0.6	59.5	88.4	70.1	<u>0.028</u>	371.5	95.2
GEOSTEER (Ours)		63.2	-7.4	2.1	69.4	<u>73.3</u>	95.8	0.018	25.7	99.4
Original		50.0	-5.4	5.2	63.8	78.2	85.2	0.097	21.2	99.2
RepE	Qwen2.5-7B	50.7	-5.4	5.6	62.5	77.3	85.1	0.109	21.9	99.1
ITI		48.8	-5.6	<u>5.8</u>	65.4	78.9	86.3	0.097	21.2	99.1
CAA		49.8	-5.4	5.1	66.3	80.0	85.8	0.049	20.3	99.3
MiMiC		48.1	-6.2	4.0	62.8	82.1	79.9	0.091	21.2	99.1
Linear-AcT		47.5	-5.9	5.3	63.6	79.3	83.9	0.084	22.1	99.2
ODESteer		<u>51.4</u>	<u>-5.4</u>	5.7	63.7	79.3	84.1	0.087	21.8	99.0
HPR		48.4	-5.9	3.8	65.6	77.9	84.3	0.046	23.5	99.1
Angular Steering		46.6	-6.2	4.7	<u>66.3</u>	85.6	80.3	0.049	21.4	99.2
Spherical Steering		17.2	-15.8	-3.0	61.2	81.7	78.6	<u>0.034</u>	125.9	97.8
GEOSTEER (Ours)		53.2	-4.6	6.2	68.1	<u>82.3</u>	85.2	0.029	22.1	99.2

cityPrompts. These results show that GEOSTEER can effectively steer model behavior toward desir-

able attributes while maintaining strong generation quality. The improvement over additive and linear

steering methods, such as CAA, MiMiC, ITI, and Linear-AcT, suggests the importance of respecting the geometry of the activation space. Since these methods typically apply a fixed Euclidean intervention, they may distort the activation representation or move it away from the model’s learned activation manifold. In contrast, GEOSTEER performs steering directly on the hypersphere, preserving the activation norm throughout the update. Compared with existing norm-preserving methods, including HPR, Angular Steering, and Spherical Steering, GEOSTEER further benefits from its adaptive geodesic trajectory. Rather than assuming a fixed transformation or a single rotation path, GEOSTEER recomputes the steering direction at each intermediate point. By dividing a large steering step into multiple smaller geodesic updates, GEOSTEER can follow a more flexible path toward the desired activation region, leading to stronger and more consistent performance across models and tasks.

4.2 Ablation Studies

We conduct ablation studies to analyze the effect of three important design choices in GEOSTEER: the number of geodesic steps, the total steering strength, and the objective function design. All ablations are evaluated on TruthfulQA using $\text{True} \times \text{Info}$ as the primary metric.

Effect of the Number of Steps. Figure 2a studies how the number of geodesic steps affects performance when the total steering strength is fixed. Increasing the number of steps, or equivalently reducing the step size, brings a modest initial gain before the performance stabilizes. This suggests that a moderate number of steps is sufficient for reliable integration, and that GEOSTEER is not sensitive to the exact step-size choice. The stable trend also shows that the objective function provides a reliable steering direction along the path. Rather than relying on one large intervention, GEOSTEER repeatedly recomputes the direction at intermediate points, allowing the activation to move more smoothly on the sphere.

Effect of Steering Strength. Figure 2b shows the effect of the total steering strength T , with the number of steps fixed to 15. Overall, GEOSTEER is not overly sensitive to small changes in T , and the performance remains strong within a reasonable range. When T is too small, the intervention is too weak to sufficiently steer the model, leading to limited gains. In contrast, when T is too large,

Table 2: Runtime analysis across different backbone models. Tokens/s measures autoregressive generation speed.

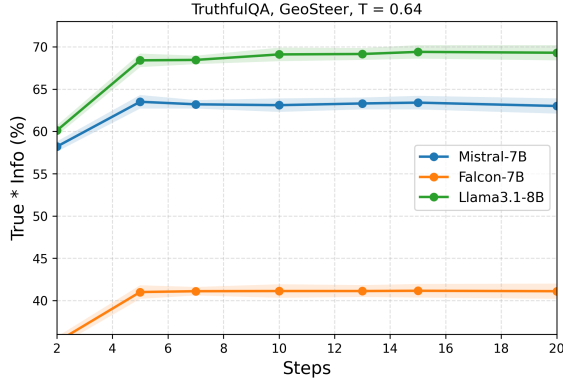
Method	Falcon-7B	Mistral-7B	LLaMA3.1-8B
Original	104.2	96.1	104.6
RepE	104.0	95.7	102.9
ITI	104.0	95.7	103.0
CAA	104.1	95.8	103.2
MiMiC	101.0	93.9	100.8
Linear-AcT	104.1	95.9	103.3
ODESteer	103.8	95.5	102.9
Angular Steering	104.1	95.7	103.0
Spherical Steering	104.1	95.6	103.1
GEOSTEER (Ours)	103.9	95.5	102.8

Table 3: Ablation study on the objective function design using LLaMA3.1-8B. We compare a linear objective against two nonlinear alternatives based on Random Fourier Features (RFF) and Polynomial Count Sketch.

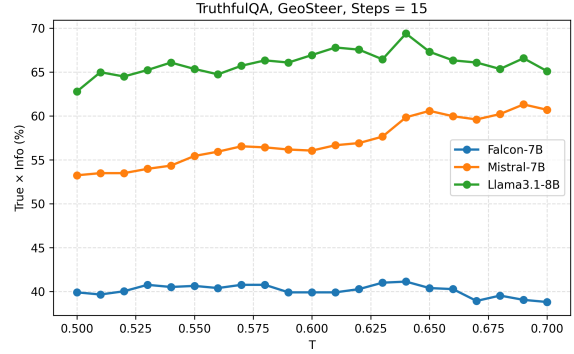
Objective Function	True $\uparrow$	Info $\uparrow$	True $\times$ Info $\uparrow$
Linear	66.2	96.2	62.4
RFF	71.4	96.0	67.8
Polynomial Count Sketch	73.3	95.8	69.4

the intervention can overly perturb the activation and degrade generation quality, which in turn reduces performance. Different models still exhibit slightly different preferred ranges. For Mistral-7B and LLaMA3.1-8B, performance improves as T increases from 0.5 and remains strong around $T = 0.64$. Falcon-7B is more stable across the tested range, with smaller performance variation. Based on these results, we set $T = 0.64$ in the main experiments, as it provides a strong and stable setting across models without requiring model-specific tuning.

Effect of Objective Function Design. Table 3 studies how the choice of objective function affects the steering performance. We compare a simple linear objective with two nonlinear alternatives: Random Fourier Features (RFF) (Rahimi and Recht, 2007) and Polynomial Count Sketch. The linear objective already achieves strong information preservation, with an Info score of 96.2, but its True score is relatively lower. This suggests that a purely linear objective can preserve the original generation quality, but is less effective at capturing the nonlinear separation between desired and undesired activation regions. Introducing nonlinear objectives improves the True score while maintaining a similar Info score. RFF increases the True score from 66.2 to 71.4, leading to a clear improvement in the combined True $\times$ Info metric.



(a) Effect of the number of geodesic steps.



(b) Effect of total steering strength T .

Figure 2: Ablation studies on TruthfulQA using $\text{True} \times \text{Info}$ as the main metric. We analyze the effect of the number of geodesic steps and the total steering strength T .

Polynomial Count Sketch performs the best among the three objectives, achieving 73.3 on True and 69.4 on $\text{True} \times \text{Info}$, while keeping Info nearly unchanged. These results indicate that modeling nonlinear relationships in the activation space provides a more effective steering direction without sacrificing informativeness.

4.3 Runtime analysis

We further evaluate the computational overhead of GEOSTEER during autoregressive generation. The experiment measures generation throughput in tokens per second on 4 NVIDIA A100 40GB GPUs across three backbone models. As shown in Table 2, the original models achieve 104.2, 96.1, and 104.6 tokens/s on Falcon-7B, Mistral-7B, and LLaMA3.1-8B, respectively. GEOSTEER achieves 103.9, 95.5, and 102.8 tokens/s on the same backbones, corresponding to relative overheads of 0.3%, 0.6%, and 1.7%, respectively. On average, GEOSTEER retains 100.7 tokens/s compared with 101.6 tokens/s for the original models, resulting in an average overhead of only 0.9%. This small decrease is expected because GEOSTEER performs iterative geodesic updates during generation. Nevertheless, the overhead remains negligible in practice and is comparable to other efficient steering methods. These results show that GEOSTEER preserves almost all of the original generation throughput, making it practical for autoregressive inference.

5 Conclusion

In this work, we introduced GEOSTEER, an optimization-based method for norm-preserving

activation steering. Unlike existing methods that rely on predefined one-step edits, GEOSTEER formulates steering as a Riemannian optimization problem and moves activations through a sequence of small geodesic steps on the activation sphere. At each step, the steering direction is adaptively recomputed from the local gradient of a learned nonlinear activation-space objective. This design better respects the geometry of normalized representations and provides a smoother and more stable way to control model behavior at inference time. Across TruthfulQA, RealToxicityPrompts, and UltraFeedback, GEOSTEER consistently improves over strong activation steering baselines. Our ablation studies further show the importance of the number of geodesic steps, the total steering strength, and the nonlinear objective function design, while the runtime analysis demonstrates that the additional cost remains modest during autoregressive generation.

6 Limitations

Although GEOSTEER provides an effective framework for norm-preserving activation steering, it has several limitations.

- **Integration with other direction-discovery methods.** Our current implementation learns the steering objective from contrastive activation examples. This makes the method simple and compatible with standard activation-steering pipelines, but it does not directly incorporate other direction-discovery mechanisms, such as sparse autoencoders (SAEs). Extending GEOSTEER to SAE-based features is non-trivial because SAE representations are

often high-dimensional, sparse, and feature-wise rather than direction-wise. It is therefore unclear how to select stable feature combinations, convert them into smooth tangent directions on the activation sphere, and perform geodesic updates efficiently without introducing noise or instability.

- **Potential risks and unintended behavior.** Activation steering is a general-purpose inference-time control mechanism. Although our experiments focus on desirable behaviors, such as improving truthfulness, reducing toxicity, and improving preference-oriented generation, the same type of method could be misused with inappropriate objectives or steering directions to induce undesirable behaviors, including toxic, biased, deceptive, or otherwise harmful outputs. In addition, preserving the activation norm does not guarantee that all unintended behavioral changes are avoided. Therefore, activation steering methods should be carefully evaluated for downstream safety, robustness, and misuse risks before deployment.

As future work, we plan to investigate stable integration of GEOSTEER with SAE-based feature discovery and to conduct broader safety evaluations that better characterize unintended behavioral changes and misuse risks across tasks and model families.

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A Appendix

A.1 LLM Usage

In this work, Large Language Models (LLMs) were used only as writing and coding assistants. Specifically, they were used to help polish the manuscript for grammar, clarity, and readability, and to draft small portions of experimental code.

All LLM-assisted content was carefully reviewed, verified, and revised by the authors. The research ideas, theoretical framework, methodology, experimental design, and analysis were developed by the authors. LLMs were not used to generate the core scientific contributions or to draw conclusions from the experimental results.

The authors take full responsibility for the correctness of the theoretical claims, the validity of the experiments, and the reported results. Any text or code produced with LLM assistance was checked to ensure correctness, originality, and compliance with ethical research standards.

A.2 Scientific Artifacts

Our experiments use only publicly available scientific artifacts, including benchmark datasets, pretrained language models, evaluation models, and baseline implementations. The datasets used in our evaluation include TruthfulQA, RealToxicityPrompts, and UltraFeedback/UltraFeedback Binarized. TruthfulQA is used for truthfulness evaluation, RealToxicityPrompts is used for toxicity-oriented generation evaluation, and UltraFeedback is used for preference-oriented evaluation. These datasets are English-language text benchmarks and are used only for research evaluation. We report aggregate results and do not release newly collected human-subject data.

We use all artifacts in accordance with their respective licenses and terms of use. TruthfulQA and RealToxicityPrompts are released under Apache-2.0, while UltraFeedback/UltraFeedback Binarized is released under the MIT license. The pretrained language models used in our experiments are also publicly available: Falcon-7B, Mistral-7B, and Qwen2.5-7B are released under Apache-2.0, while LLaMA3.1-8B is released under the LLaMA 3.1 Community License. We use these models only as research artifacts for controlled evaluation of activation-steering methods.

Our use of these artifacts is consistent with their intended research and evaluation purposes. We do not redistribute modified versions of the datasets or

pretrained models as part of this work.

We use publicly available benchmark datasets and do not collect new human-subject data. Some datasets used in our experiments may contain sensitive or offensive text by construction, especially RealToxicityPrompts, which is designed for evaluating toxic language generation. We use these datasets only for research evaluation, do not redistribute modified versions of the data, and report aggregate evaluation results rather than individual examples. We do not release any newly collected personally identifying information.

A.3 PCS Hyperparameters

As described in 3.2, we use Polynomial Count Sketch (Pham and Pagh, 2013) to construct randomized polynomial features. This provides an efficient approximation to the polynomial kernel

$$K(x, y) = (\gamma x^\top y + c_0)^d, \quad (22)$$

where γ controls the scale of the inner product, c_0 is the constant offset, and d denotes the polynomial degree. The approximation also depends on the number of random features, denoted by N_{poly} . We set $\gamma = 0.1$, $c_0 = 1.0$, $d = 2$, and $N_{\text{poly}} = 8000$ in all experiments. We found this configuration to be stable across different datasets and base models.

A.4 Norm Preservation of GeoSteer

We show that the GeoSteer update preserves the unit-norm constraint. At step t , assume that the current activation direction lies on the unit sphere:

$$\|z^{(t)}\|_2 = 1. \quad (23)$$

The update direction $u^{(t)}$ is obtained by projecting the gradient onto the tangent space of the sphere and normalizing it. Therefore,

$$\|u^{(t)}\|_2 = 1, \quad \left(z^{(t)}\right)^\top u^{(t)} = 0. \quad (24)$$

GeoSteer updates the activation direction as

$$z^{(t+1)} = \cos(\eta)z^{(t)} + \sin(\eta)u^{(t)}. \quad (25)$$

To verify that the updated point remains on the sphere, we compute its squared norm:

$$\left\|z^{(t+1)}\right\|_2^2 = \left\|\cos(\eta)z^{(t)} + \sin(\eta)u^{(t)}\right\|_2^2. \quad (26)$$

Expanding the squared norm gives

$$\begin{aligned} \|z^{(t+1)}\|_2^2 &= \cos^2(\eta) \|z^{(t)}\|_2^2 + \sin^2(\eta) \|u^{(t)}\|_2^2 \\ &\quad + 2 \cos(\eta) \sin(\eta) (z^{(t)})^\top u^{(t)}. \end{aligned} \quad (27)$$

Since $z^{(t)}$ and $u^{(t)}$ are unit vectors and are orthogonal, we obtain

$$\begin{aligned} \|z^{(t+1)}\|_2^2 &= \cos^2(\eta) + \sin^2(\eta) \\ &= 1. \end{aligned} \quad (28)$$

Thus,

$$z^{(t+1)} \in \mathbb{S}^{d-1}. \quad (29)$$

Since the initial point $z^{(0)} = h/\|h\|_2$ is on the unit sphere, the same argument applies at every step. Therefore, all intermediate points remain on the sphere, and the final reconstructed activation

$$\hat{h} = \|h\|_2 z^{(K)} \quad (30)$$

preserves the original activation norm.

A.5 Steering Model Details

We conduct experiments using the following open-source language models:

- Falcon-7B: `tiiuae/falcon-7b`;
- Mistral-7B: `mistralai/Mistral-7B-v0.3`;
- LLaMA3.1-8B: `meta-llama/Llama-3.1-8B`;
- Qwen2.5-7B: `Qwen/Qwen2.5-7B`.

For all models and tasks, we use the same generation configuration: temperature is set to 0.7, top- p to 0.9, and repetition penalty to 1.1.

A.6 Steering Position

For a fair comparison, we apply GEOSTEER and all baselines at the same residual-stream position within each model. Steering is applied to all newly generated tokens during decoding. To choose the steering layer, we first run CAA (Rimsky et al., 2024) on TruthfulQA across all layers of each model and evaluate performance using the True×Info metric. The layer-wise results are shown in Figure 3.

Based on this analysis, we use layer 15 for Falcon-7B, layer 16 for Mistral-7B, layer 14 for LLaMA3.1-8B, and layer 14 for Qwen2.5-7B. We

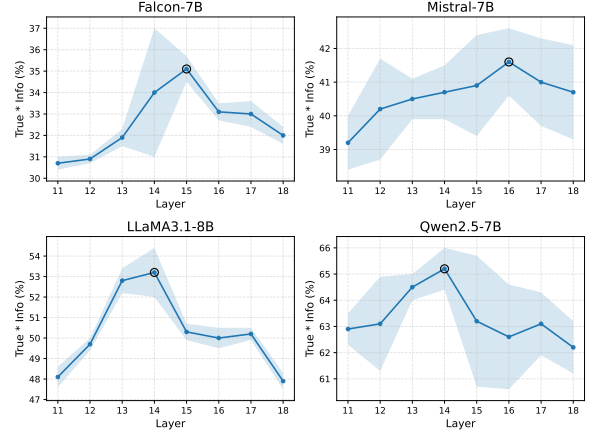


Figure 3: Layer-wise steering performance on TruthfulQA using CAA for layer selection. We report True×Info across layers 11–18 for each base model.

use CAA only as a layer-selection tool, so that all methods are evaluated under the same steering position rather than tuning the layer separately for each method.

A.7 Datasets

UltraFeedback. We use the UltraFeedback Binarized dataset, where each prompt is associated with a preferred response and a rejected response. From this dataset, we construct 10k training pairs, 500 validation pairs, and 500 test prompts under three random seeds. For evaluation, we score model outputs using Skywork-Reward-V2-LLaMA-3.1-8B. We report the average reward score, denoted as RM-mean, the 90th percentile reward score, denoted as RMP90, and the reward-model win rate relative to the baseline model.

Reward-Model Win Rate. Given a set of prompts $\{x_i\}_{i=1}^N$ and two systems A and B , let s_i^A and s_i^B denote their reward-model scores on the same prompt. Following (Lambert et al., 2025), we define the win rate of system A over system B as:

$$\begin{aligned} \text{Win}(A, B) &= \frac{1}{N} \sum_{i=1}^N \left[\mathbb{I}(s_i^A > s_i^B) \right. \\ &\quad \left. + \frac{1}{2} \mathbb{I}(s_i^A = s_i^B) \right]. \end{aligned} \quad (31)$$

A win rate of 0.5 indicates parity with the baseline, while values above 0.5 indicate that system A is preferred by the reward model more often than system B . Ties are counted as half wins.

TruthfulQA. For TruthfulQA, we follow the generation-based evaluation setting used by Li et

al. (2023). The benchmark contains 817 questions, which are expanded into 5,918 question–answer pairs. We use 40% of the data for training and 10% for validation to select hyperparameters. We then conduct two-fold cross validation so that all TruthfulQA questions are included in the test split. The original TruthfulQA benchmark used two fine-tuned GPT-3 models as truthfulness and informativeness judges. Since these models are no longer publicly available, we instead use `allenai/truthfulqa-truth-judge-llama2-7B`, `allenai/truthfulqa-info-judge-llama2-7B` to evaluate truthfulness and informativeness, respectively.

RealToxicityPrompts. For detoxification, we use the Jigsaw Unintended Bias in Toxicity Classification dataset for training and RealToxicityPrompts (Gehman et al., 2020) for testing. Specifically, we uniformly sample 10k sentences from the Jigsaw dataset according to toxicity scores, resulting in 5k toxic and 5k benign training examples. For evaluation, we select 500 toxic prompts from RealToxicityPrompts as inputs to the language models. We use `unitary/toxic-bert` to measure the toxicity of generated continuations. In addition, we report perplexity and Dist- n scores to assess generation quality and diversity.

Activation Collection. For TruthfulQA and Ultra-Feedback, each training example contains a prompt paired with a positive and a negative response. We construct the model input by concatenating the prompt with each corresponding response and then forward the full sequence through the LLM. For the detoxification task, the Jigsaw dataset does not provide question–answer pairs; therefore, we directly use the toxic and non-toxic prompts as inputs for activation extraction. Following common practice in activation steering (Wehner et al., 2025), we extract hidden states at the final token position of each input sequence to obtain positive and negative activations. This design aligns with the autoregressive decoding process, where steering is applied when predicting newly generated tokens.